

Practical 2

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lab-03-34@lab-03-34-OptiPlex-3280-AIO:~$ vim practical2_1275.l
lab-03-34@lab-03-34-OptiPlex-3280-AIO:~$ lex practical2_1275.l
lab-03-34@lab-03-34-OptiPlex-3280-AIO:~$ cc lex.yy.c -ll
lab-03-34@lab-03-34-OptiPlex-3280-AIO:~$ ./a.out
Enter C program (Press Ctrl+D to end):
#include<stdio.h>

// This is a comment
int main()
{
    int a = 10;
    float b = 20.5;

    /* Multi-line
       comment */

    if(a > 5)
        printf("Hello");

    return 0;
}

----- Result -----
Lines      : 16
Spaces     : 10
Words      : 14
Characters : 183
Keywords   : 5
Identifiers : 9
Comments   : 2
lab-03-34@lab-03-34-OptiPlex-3280-AIO:~$
```

```
int main()
{
    printf("Enter C program (Press Ctrl+D to end):\n");
    yylex();

    printf("\n----- Result ----- \n");
    printf("Lines      : %d\n", lines);
    printf("Spaces     : %d\n", spaces);
    printf("Words      : %d\n", words);
    printf("Characters  : %d\n", chars);
    printf("Keywords   : %d\n", keywords);
    printf("Identifiers : %d\n", identifiers);
    printf("Comments   : %d\n", comments);

    return 0;
}

int yywrap()
{
    return 1;
}
```

```

%{
#include <stdio.h>

int lines = 1, spaces = 0, words = 0, chars = 0;
int keywords = 0, identifiers = 0, comments = 0;

char *key[] = {
    "int", "float", "char", "double", "if", "else", "while", "for",
    "return", "void", "break", "continue", "switch", "case", "default",
    "do", "long", "short", "signed", "unsigned", "const", "static", "struct"
};

int isKeyword(char *str)
{
    int i;
    for(i = 0; i < 23; i++)
    {
        if(strcmp(str, key[i]) == 0)
            return 1;
    }
    return 0;
}
}%

LETTER      [a-zA-Z_]
DIGIT       [0-9]
ID          {LETTER}({LETTER}|{DIGIT})*

%%

"//".*      { comments++; chars += yyleng; }

"/*"([^\*]|\\*+([^\*/])*\*+/" { comments++; chars += yyleng; }

\n          { lines++; chars++; }

[ ]         { spaces++; chars++; }

[ \t]+      { chars += yyleng; }

{ID}        {
                words++;
                chars += yyleng;
                if(isKeyword(yytext))
                    keywords++;
            else

```